COMPUTER ENGINEERING

QUEZON CITY

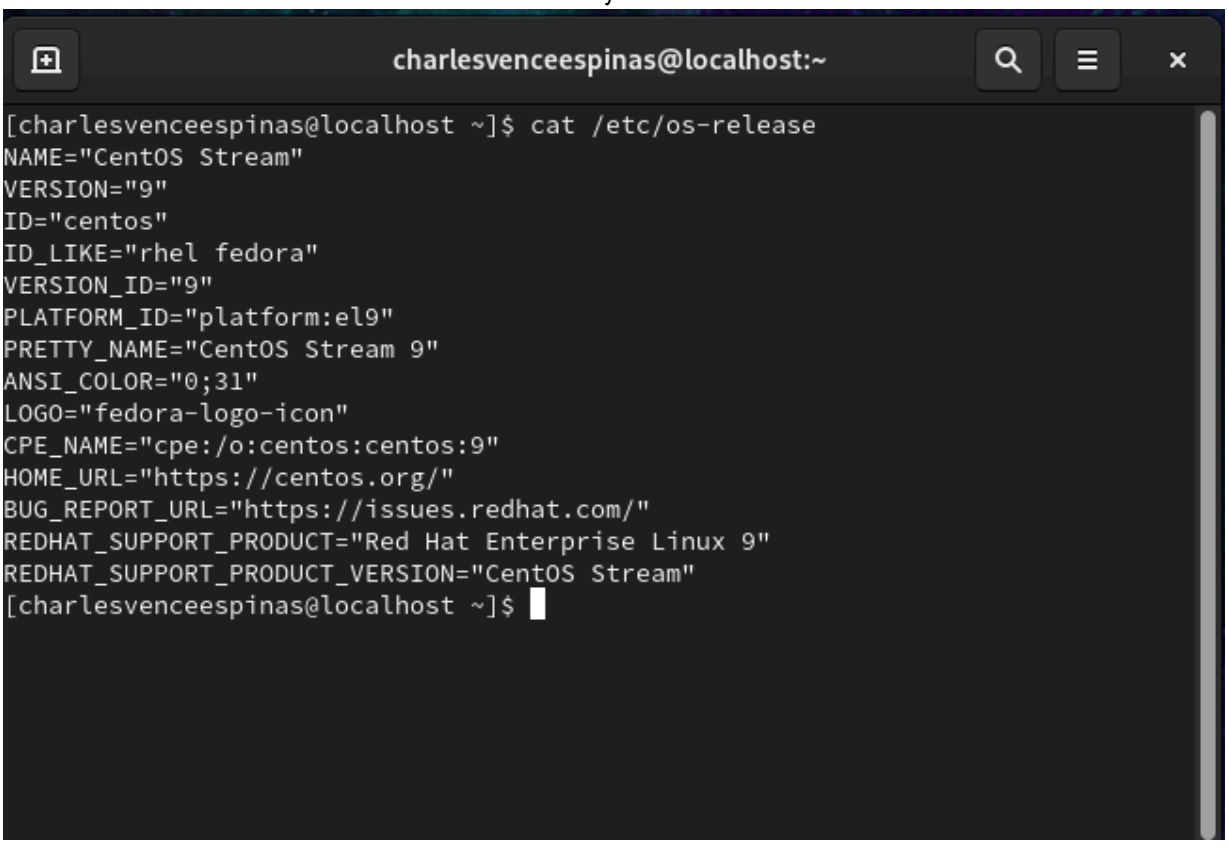
COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07/23/2026	**DATE SUBMITTED:**	07/23/2026
NAME:	Charles Vence Espinas	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 3 **Install SSH server on CentOS or RHEL 8**

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:

$ dnf install openssh-server

```
[charlesvenceespinas@localhost ~]$ sudo dnf install openssh-server
Last metadata expiration check: 0:08:23 ago on Thu 23 Jul 2026 10:17:43 AM PST.
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Start the *sshd* daemon and set to start after reboot:
   $ systemctl start sshd

   $ systemctl enable sshd

```
[charlesvenceespinas@localhost ~]$ systemctl start sshd
[charlesvenceespinas@localhost ~]$ systemctl enable  sshd
[charlesvenceespinas@localhost ~]$ systemctl status sshd
```

3. Confirm that the sshd daemon is up and running:
   $ systemctl status sshd

```
[charlesvenceespinas@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-07-23 10:19:06 PST; 7min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 926 (sshd)
      Tasks: 1 (limit: 10500)
     Memory: 2.1M (peak: 2.4M)
        CPU: 17ms
     CGroup: /system.slice/sshd.service
             └─926 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 10:19:06 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Jul 23 10:19:06 localhost.localdomain sshd[926]: Server listening on 0.0.0.0 port 22.
Jul 23 10:19:06 localhost.localdomain sshd[926]: Server listening on :: port 22.
Jul 23 10:19:06 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

4. Open the SSH port 22 to allow incoming traffic:
   $ firewall-cmd --zone=public --permanent --add-service=ssh

   $ firewall-cmd --reload

```
[charlesvenceespinas@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[charlesvenceespinas@localhost ~]$ firewall-cmd --reload
success
```

5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
   $ systemctl reload sshd

```
[charlesvenceespinas@localhost ~]$ ls -l /etc/ssh/sshd_config
-rw-------. 1 root root 3674 Jul 14 21:18 /etc/ssh/sshd_config
[charlesvenceespinas@localhost ~]$ systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.

```
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ ls ~/.ssh/id_rsa.pub
ls: cannot access '/home/charles-vence-espinas/.ssh/id_rsa.pub': No such file or directory
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/charles-vence-espinas/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/charles-vence-espinas/.ssh/id_rsa
Your public key has been saved in /home/charles-vence-espinas/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:tzQ0aHYyzTippCqBrt06Q37BIS4YHEmFlSaLycyVqcg charles-vence-espinas@charles-vence-espinas-VirtualBox
The key's randomart image is:
+---[RSA 4096]----+
|..=o+            |
| = *     *       |
|O.B   . X *      |
|*E. .o + * .     |
|+o o... S +      |
|+.o.o    o o     |
|.=.  .    .      |
|.o+..            |
|. o=.            |
+----[SHA256]-----+
```

Generate rsa key in local machine

```
[charlesvenceespinas@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f2:be:b9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
       valid_lft 526sec preferred_lft 526sec
    inet6 fe80::a00:27ff:fef2:beb9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[charlesvenceespinas@localhost ~]$
```

Getting the ip address of centOS machine

2. Using the command *ssh-copy-id*, connect your local machine to CentOS.

```
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ ssh-copy-id charlesvenceespinas@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:jDqbbHTbBZEnAGPZ5H1SDTyPak5S+FllmL0u7ZSMJms.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
charlesvenceespinas@192.168.56.102's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'charlesvenceespinas@192.168.56.102'"
and check to make sure that only the key(s) you wanted were added.
```

3. On CentOS, verify that you have the *authorized_keys*.

```
charlesvenceespinas@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQC+qpBuACGwu//DWphM9BoPuOZWss8v4gh5vuw4r4KY6nACbZ2n4s/kqfQPwelyfXFk15NJfvbFd+/qqtQFcwJM32nMut3QODj6gPNFnMBlJ1al3HpQ353ArT
P1K3Y2+LBxjA2oDMkK9nbfglkYBW4t2G6qe1wCmIAGkc3HaCih5P1pQ1IoIQiDMT/MPJz9noe3tM8KeP6sUyBi512+aoTdpQmUNpzV+g0wOMTfEMzja22TE2ZSzNyILnaSCssbZoqlemgS3SIQs2aFSPKIrU98
29HjiMEkM5Gur38kVPTdDXYtolscN2ojQ1+h0B2GCRSEAWlW1M/Y2GUVqrLStQ2oM+94/aRVT5KthAzhKtVH5GIqzriOgaz8Ks9ZRzEGPaAagHVow8g996VziMNjSlnSiDWhziYoNtKSUiNCPvQywEFPgYguh
Hb9rt9MKf6BPJJ+6OTEZRA0kdqbwDUjvWWtA/wtlyb6W1SrGdP8710BzUZDUBCmllFOYQUjY5HLUhZzD7qRf/QPELeEu9OXJi0/Z4yBI1pp38pCyPxsEjff5UEUQf/NUKfkBC+T3fTA/2lczetrwGUBPLc74T4
i4Vw19x57745reGZGE6xIMCvl9pELdwoNmHK0oFX/7i4Fau2AgOzEFnRfi3Ws70L2bISInz6yHMBHsuO4tdkB0qL+w== charles-vence-espinas@charles-vence-espinas-VirtualBox
[charlesvenceespinas@localhost ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
charles-vence-espinas@charles-vence-espinas-VirtualBox:~$ ssh charlesvenceespinas@192.168.56.102
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 18:43:16 2026
```

2. Show evidence that you are connected.

```
[charlesvenceespinas@localhost ~]$ whoami
charlesvenceespinas
[charlesvenceespinas@localhost ~]$ cat /etc/redhat-release
CentOS Stream release 9
[charlesvenceespinas@localhost ~]$ cat /etc/os-release
NAME="CentOS Stream"
VERSION="9"
ID="centos"
ID_LIKE="rhel fedora"
VERSION_ID="9"
PLATFORM_ID="platform:el9"
PRETTY_NAME="CentOS Stream 9"
ANSI_COLOR="0;31"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:centos:centos:9"
HOME_URL="https://centos.org/"
BUG_REPORT_URL="https://issues.redhat.com/"
REDHAT_SUPPORT_PRODUCT="Red Hat Enterprise Linux 9"
REDHAT_SUPPORT_PRODUCT_VERSION="CentOS Stream"
[charlesvenceespinas@localhost ~]$ who
charlesvenceespinas tty2         2026-07-23 18:43 (local)
charlesvenceespinas pts/1        2026-07-23 18:48 (192.168.56.101)
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. **What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?**
   - I think the things that we should look for in choosing the best distribution between Debian and Red Hat Linux distributions are the purpose of use, hardware compatibility, available software, security, package management and ease of use. We should also consider which distribution is more suitable for our needs and experience as users.

2. What are the main differences between Debian and Red Hat Linux distributions?

- The main differences between Debian and Red Hat Linux distributions are their package management systems, package formats and the distribution that are based on them. Debian uses the DEB package format and APT as its package manager while the Red Hat uses the RPM package format and DNF or YUM as its package manager. Denial is also known for being a free and community driven distribution while the Red Hat Linux is mainly focused on the enterprise use and provides commercial support. Both distributions are reliable but they have different systems and are commonly used for different purposes.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use this tool once in this activity to help me figure out what command I should use in the given scenario.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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